

Solve at least two of the following problems. Post your solutions to the Discussions thread on Canvas by 11:59pm on the due date. It is fine if multiple people post solutions to the same problem. Also, this is not a race! The discussion thread supports the use of LaTeX, and using this is the preferred method for sharing your work. You can also upload a pdf or other file if needed. You may collaborate with others—this is encouraged!—but please note your collaboration on any solutions you post. Similarly, note if you use any outside resources. If you have any clarifying questions about a problem, please ask the the author about it in a polite and generous fashion.

Agustin Valenzuela Pidal: Solve the following system of equations of a subspace in $\mathbb{R}^3$:

$$-4x_1 + 3x_2 - 16x_3 + x_4 = 0$$

$$8x_1 - 2x_2 + 8x_3 = 0$$

$$3x_1 - x_2 + 2x_3 + 2x_4 = 0$$

Determine its rank and state how many free variables it has, if any.

Alan Rodal:

Find the inverse of the following matrix A and compute the product AA^{-1} to confirm it equals I .

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

Andrew Mark Welch:

1. Compute the volume of the parallelepiped defined by the three vectors

$$v_1 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, v_3 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}.$$

2. Suppose v_1 gets scaled by 2 and v_3 gets scaled by 3. What effect does this have on the volume of the parallelepiped?

Arina Oberoi: Give an example of a linearly independent vector space for the following requirements. If this isn't possible, explain why.

- a) Four vectors in the $\mathbb{R}^3$ space
- b) The zero vector
- c) Two vectors in the $\mathbb{R}^3$ space
- d) Four vectors in the $\mathbb{R}^5$ space where the last row of each is 0

Compose a linearly independent vector space, then change something to make it linearly dependent.

Arjun Sohur:

Consider $n \times n$ matrix A where $a_{ij} = 0$ for all $i = j$ on $1 \leq i, j \leq n$. Is it possible for A to be invertible?

Arya Vohra:

Let V and W be vector spaces over the same field $\mathbb{F}$, and let $T : V \rightarrow W$ be a linear map. Suppose $\dim(V) = n$ and $\dim(W) = m$. Prove that T is injective if and only if its null space $\text{null}(T)$ is trivial.

Bonnie Ko:

Let V and W be vector spaces, and let $\alpha : V \rightarrow W$ be a linear transformation. Show that the image of α , denoted by $\text{Im}(\alpha)$, is a subspace of W .

Brian Li:

Give a basis for the plane $2x + 3y + z = 0$ and one for the line $2x = y$. Is the union of the set of these 2 bases a basis for $\mathbb{R}^3$? Justify.

Claire Kim: Is there a 3×4 matrix A such that AA^T would be invertible?

Emilio Pelaez:

Given the matrices $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, find a gate A such that $AXA = Z$ and $AZA = X$.

Emmanuel Abayev:

Show that if A is any $n \times n$ symmetric matrix, then $I_n * A * 2A^2 * 3A^3$ is also symmetric.

Evan Mathura:

Let us have a system of vectors $v_1, v_2, v_3, v_4, v_5 \in \mathbb{F}^3$. Does this system of vectors form a basis of $\mathbb{F}^3$? Why or why not?

Felix Farb:

When I did math competitions in high school, we were taught to use something called the shoelace formula to quickly find the area of polygons in the coordinate plane:

<https://www.tuitionkenneth.com/a-maths-area-shoelace-method>

We proved this algebraically in high school using triangles, but prove it here using determinants.

Here's another one that I fully came up with:

Prove that $\det(A^{-1}) = 1/(\det(A))$

Frank Ragone:

Find the inverse of $\begin{pmatrix} 1 & 2i \\ 2 & 6i \end{pmatrix}$.

George Lord:

Digital grayscale (black and white) image files *could* be represented as matrices of integers between 0 and 128, where each decimal at some row and column in the matrix represents the gray color of the pixel at the same row and column in the displayed image, with 0 being completely white, 128 being completely black, and values closer to 0 being more light-gray and those closer to 128 being more dark-gray.

These integers would be represented as numbers in binary (base 2) with 8 places, with numbers such as 128_{10} being represented as 10000000_2 , 0_{10} represented as 00000000_2 , or 73_{10} being represented as 01001001_2 . Each place in these numbers takes up a “bit” of memory, which corresponds to a very small electronic circuit in your computer’s memory being “off” for 0 or “on” for 1. A group of 8 bits is called a byte, and is used as the standard unit of computer storage. As such, each number in our theoretical grayscale image would take 8 bits of memory to store, with our entire image taking approximately $m \cdot n \cdot 8$ bits to store, or $m \cdot n$ bytes.

Usually, digital image files are “compressed” to reduce the amount of computer storage they consume while still resembling the original image (although small discrepancies can be seen when zooming in).

Take some arbitrary $m \times n$ matrix A of integers between 0 and 128, corresponding to a grayscale image with a width of n pixels and a height of m pixels. First, calculate the amount of memory needed to store this grayscale image under our assumptions. Then, use your knowledge of matrix manipulation and linear algebra to devise an algorithm to “compress” this image into a matrix/group of matrices that consume less total memory than the original, while being able to be mathematically manipulated into a matrix resembling the original image. There are many ways to do this, and the desired quality of compression is subjective. This can be done using plain mathematics, programming, or both, but will undoubtedly (and should) use some of the knowledge you have gained in MATH 20250.

Han Zhang:

Let $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 4 & -1 \end{pmatrix}$

Find the general solution to $AZ = 0$.

Illia Voloshyn:

Short version: Prove that the perspective projection is not linear. Long version: given transformation $(x, y, z)^T \rightarrow (x/(1 - z/d), y/(1 - z/d), 0)^T$ for $d \neq z$ and $d \neq 0$, show that this transformation is not linear.

Jacky Chen:

In $M_{3 \times 3}^{\mathbb{F}}$, define $W_1 = \{A \in M_{3 \times 3}^{\mathbb{F}} | A_{ij} = 0 \text{ whenever } i < j\}$ and $W_2 = \{B \in M_{3 \times 3}^{\mathbb{F}} | B = B^T\}$. Show that $W_1 \oplus W_2 = M_{3 \times 3}^{\mathbb{F}}$.

Judson Buchko:

Find a linear transformation from $(\mathbb{Z}/5\mathbb{Z})^2$ to $(\mathbb{Z}/10\mathbb{Z})^2$, $A : (\mathbb{Z}/5\mathbb{Z})^2 \rightarrow (\mathbb{Z}/10\mathbb{Z})^2$. Can A be right invertible, can A be left invertible? If so, provide an example of the matrix and its right/left inverse, if not, explain why.

Juliet Oliver:

See the last page of the document.

Justin Tarquino:

Find the inverse of the following matrix. If it doesn't exist, show why.

$$\begin{pmatrix} 2 & i-1 & -2 \\ 6 & 3i & 6 \\ i & -.5 & i \end{pmatrix}$$

Justin Wong:

Is a transformation from $\mathbb{F}^5$ to $\mathbb{F}^2$ such that $\text{null}\{x_1, x_2, x_3, x_4, x_5\}$ are elements of $\mathbb{F}^5$, and $x_1 = x_3 = x_5$ and $2x_2 = 4x_4$, surjective? If not why?

Khush Jammu:

Let A be an $n \times n$ matrix over a field F , and let $v_1, v_2, \dots, v_n$ be a set of vectors in a vector space V over F . Suppose that the set $Av_1, Av_2, \dots, Av_n$ is linearly independent. Consider the following transformation $T : V \rightarrow V$, defined by $T(v) = Av$ for each $v \in V$.

Prove or disprove the following statement: The linear transformation T is always injective (one-to-one). Provide a counterexample if the statement is false.

Maanav Kaistha:

Define the transformation as the following: $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where $T(v) = 0$ if $v = 0$ or if either

of the coordinates of v are irrational; and $T(v) = (1/q, 1/m)$ if both coordinates of v are rational and are p/q and n/m in their most reduced form, respectively.

A) Is this a linear transformation? Where would it send $(5/10, 2/5)^T$ to? B) Is there an inverse to this? If so, write it.

Mark O'Shea:

i) Give a few examples of a matrix that is injective but not surjective. ii) Give a few examples of a matrix that is surjective but not injective. iii) Give a few examples of a matrix that is bijective. iv) What is the general condition for a matrix to be injective but not surjective? Surjective but not injective? Bijective?

Marten Garicano:

Can a 2×2 matrix be invertible if its determinant is 0? What about a 3×3 matrix? And $n \times n$? Prove that any matrix is invertible if and only if its determinant is not 0.

Maryam Contractor:

Would the following be a subspace of $\mathbb{R}^2$? - The set of vectors $[0, p/q]$ such that p, q are elements in the natural numbers and the greatest common denominator of p and q is 1 (i.e., p/q is an element of $\mathbb{Q}$ in canonical form.)

Max Zhalilo:

Prove that the set of continuous real functions on the closed interval, $[0, 1]$ is a subspace of F on $[0, 1]$.

Michael Clancy:

Given the following Matrix, A :

$$\begin{pmatrix} 0 & 2 & 2 & 7 \\ 4 & 5 & 3 & 1 \\ 7 & 8 & 8 & a \\ 3 & 4 & 2 & -3 \end{pmatrix}$$

Determine the value of a , such that the Reduced Row Echelon form of Matrix A represents the identity matrix of $\mathbb{R}^4$.

Mikhail Gebeyehu:

Write a transformation A such that $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and A reflects a vector over the line $x_1 = -5x_2$. Is this matrix invertible? If so, can you find A^{-1} ?

Myra Singh:

Does there exist a matrix A such that $A^2 = -A$? If so, find one. If not, prove that it is not possible.

Neel Maheshwari:

Let $A = \begin{bmatrix} 1 & -5 & -2 \\ 1 & 3 & 1 \\ -1 & 4 & 2 \end{bmatrix}$. Calculate the left-inverse and the right-inverse of the matrix to see whether A^{-1} exists, implying A is invertible.

Nikita Rodin:

For every $N > 1$, is there $N \times N$ matrix such that $A^2 = B$, $B^2 = A$, $A \neq B$ and has no 0-entries.

Parth Wokhlu:

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that first reflects points through $y = 3x$ and then $y = -2x + 5$. Find the matrix that represents T .

Richard Chen:

Define an isomorphism between $\mathbb{F}^{n+1}$ and $P_n(\mathbb{F})$ and show that it is an isomorphism.

Riley Fisher:

Given the following coordinates on a graph: $\{(2, 2), (-5, 5), (-2, -2), (3, -3)\}$, define three transformations for three different coordinates that form a square in the first quadrant if the coordinates were connected to one another.

Shivant Krishnan:

Find the determinant of the matrix

$$\begin{pmatrix} 1 & 2 & 5 & 7 & 10 \\ 1 & 2 & 3 & 6 & 7 \\ 1 & 1 & 3 & 5 & 5 \\ 1 & 1 & 2 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Simon Seignourel:

How many invertible 3×3 matrices over $\mathbb{Z}/2\mathbb{Z}$ are there? (3×3 matrices where the entries can only be 0 and 1.)

Syris Shao:

If A is a square matrix and $A^2 = 0$, then $A = 0$. Is this true?

Yazid Badawy:

Find the inverse of the matrix:
$$\begin{pmatrix} 4 & 8 & 7 & 3 \\ 5 & 8 & 9 & 4 \\ 3 & 5 & 12 & 12 \\ 6 & 7 & 3 & 16 \end{pmatrix}$$

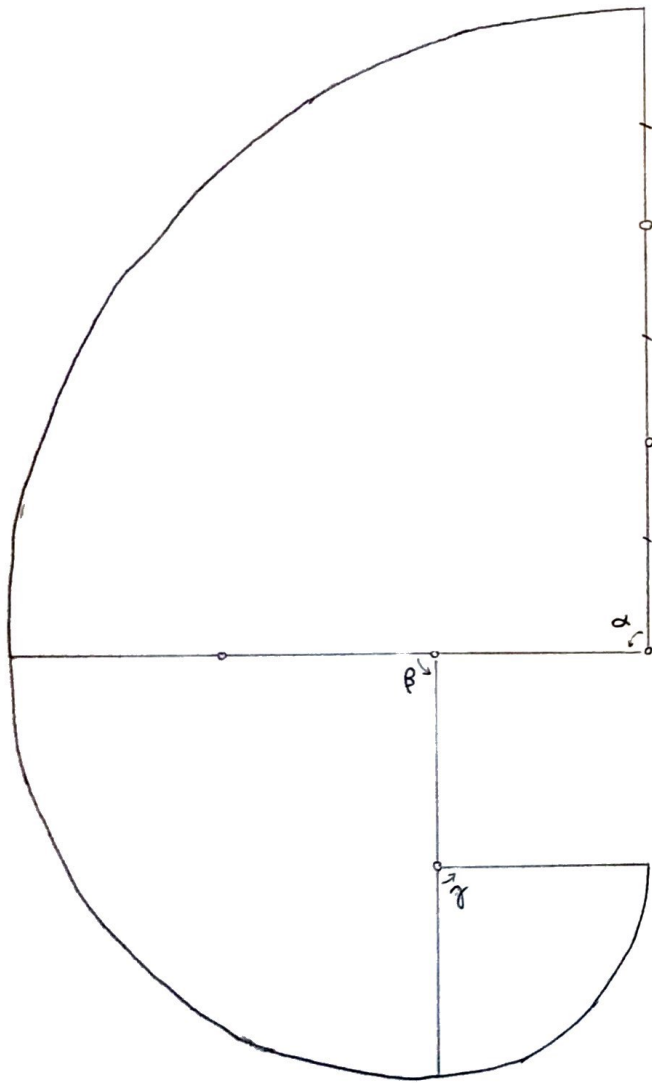
Zane Miller:

Prove the following three statements:

For all $A \in \mathcal{M}_{n \times n}^{\mathbb{F}}$ and $\alpha \in \mathbb{F}$, $\det(\alpha A) = \alpha^n \det(A)$.

For all $A \in \mathcal{M}_{n \times n}^{\mathbb{F}}$ and $\alpha \in \mathbb{F}$, $\det(\alpha A) = \alpha \det(A) \iff \det(A) = 0, \alpha = 0$, or $\alpha^{n-1} = 1$.

For all $B \in \mathcal{M}_{n \times n}^{\mathbb{C}}$ and $\beta \in \mathbb{C}$, $\det(\beta B) = \beta \det(B) \iff \det(B) = 0, \beta = 0$, or $\beta = e^{2\pi i k}$ for some $k \in \mathbb{Z}$.



Does this span $\mathbb{R}^2$?

3 segments of equal length, scalable by elements in $\mathbb{R}$

connected by 2 joints and 1 fixed at origin, each able to rotate max $\frac{\pi}{2}$ from starting position

each vector can be described as $r \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$, $r \in \mathbb{R}$, $0 \leq \alpha, \beta, \gamma \leq \frac{\pi}{2}$
where each segment has length r .

For $r < 0$, think of scaling the "starting vector", $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, by $|r|$,
flipping it across the "x-axis" and then performing the rotations
given by α, β, γ